Exercise 06: Relational database design and normal forms

Task 1*: Normal forms

Given the following relations R and S in the first normal form with functional dependencies F :

$$\begin{aligned} R &= (A,B,C,D,E,F)\ mit \\ A,B &\rightarrow C,D,E \\ D &\rightarrow F \\ A,B,C &\rightarrow D,E \end{aligned} \qquad \begin{aligned} S &= (V,W,X,Y,Z)\ mit \\ V &\rightarrow W,X,Y,Z \\ W,Z &\rightarrow V,X,Y \\ Y &\rightarrow Z. \end{aligned}$$

1. First, use the `COVER` algorithm to simplify the set of functional dependencies.
2. Determine the keys of the relations.
3. Do the relations correspond with the second normal form (2NF)?
4. Do the relations correspond with the third normal form (3NF)?
5. Do the relations correspond with the Boyce-Codd normal form (BCNF)?

Task 2: Support using ChatGPT

Feed this pdf to ChatGPT or other language models and let the AI argue why R is not in BCNF and let it generate the keys. Does this match your result?

Task 3: Some more Normal forms

Given the following relations Y in the first normal form with functional dependencies F :

$$\begin{aligned} Y &= (A,P,H,R,O,D,I,T,E)\ with \\ R &\rightarrow O \\ O &\rightarrow A,H,P \\ O,P &\rightarrow D,R \\ H,P &\rightarrow P \\ H,P,R &\rightarrow D \end{aligned}$$

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Task 4: Some more Normal forms

1. Given the following relations W' in the first normal form with functional dependencies W :

$$\begin{aligned} W' &= (A,B,C,D)\ with \\ A,B &\rightarrow C \\ B &\rightarrow D \end{aligned}$$

Show that W' does not correspond with the second normal form.

2. Given the following relations X' in the first normal form with functional dependencies X :

$$\begin{aligned} X' &= (A,B,C,D)\ with \\ A,B,C &\rightarrow D \\ B,C &\rightarrow A \end{aligned}$$

Show that X' corresponds with the third normal form.

3. Given the following relations V' in the first normal form with functional dependencies V :

$$\begin{aligned} V' &= (A,B,C,D,E)\ with \\ C,D,E &\rightarrow A,C \\ A,E &\rightarrow B,D \\ C,D &\rightarrow E \end{aligned}$$

Show that V' does not corresponds with the third normal form.

4. Given the following relations Y' in the first normal form with functional dependencies Y :

$$\begin{aligned} Y' &= (A,B,C,D,E,F)\ with \\ A &\rightarrow B,C \\ C &\rightarrow D \\ E &\rightarrow F \end{aligned}$$

Show that Y' does not correspond with the third normal form.

5. Given the following relations Z' in the first normal form with functional dependencies Z:

$$\begin{aligned} Z' &= (A,B,C,D)\ with \\ A,B,C &\rightarrow D \\ A,B &\rightarrow C,D \\ C &\rightarrow A \end{aligned}$$

Show that Z' does not correspond with the Boyes-Codd normal form.